# 云计算实验二报告

作业 1：

```
[root@10-23-126-118 ~]# docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

```
[root@10-23-126-118 ~]# docker pull busybox
Using default tag: latest
latest: Pulling from library/busybox
8e674ad76dce: Pull complete
Digest: sha256:bf510723d2cd2d4e3f5ce7e93bf1e52c8fd76831995ac3bd3f90ecc866643aff
Status: Downloaded newer image for busybox:latest
docker.io/library/busybox:latest
[root@10-23-126-118 ~]# docker images
REPOSITORY     TAG       IMAGE ID       CREATED        SIZE
hello-world    latest    9c7a54a9a43c   4 months ago   13.3kB
busybox        latest    e4db68de4ff2   4 years ago    1.22MB
[root@10-23-126-118 ~]# docker run busybox
[root@10-23-126-118 ~]# docker run busybox echo "hello from busybox"
hello from busybox
```

作业 2:

```
[root@10-23-126-118 ~]# docker pull centos
Using default tag: latest
latest: Pulling from library/centos
a02a4930cb5d: Pull complete
Digest: sha256:365fc7f33107869dfcf2b3ba220ce0aa42e16d3f8e8b3c21d72af1ee622f0cf0
Status: Downloaded newer image for centos:latest
docker.io/library/centos:latest
[root@10-23-126-118 ~]# docker run centos
[root@10-23-126-118 ~]# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[root@10-23-126-118 ~]# docker run -it centos /bin/bash
[root@77fc518c2de7 /]# docker ps
bash: docker: command not found
```

```
[root@77fc518c2de7 /]# lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Byte Order:            Little Endian
CPU(s):                1
On-line CPU(s) list:   0
Thread(s) per core:    1
Core(s) per socket:    1
Socket(s):             1
NUMA node(s):          1
Vendor ID:             GenuineIntel
CPU family:            6
Model:                 85
Model name:            Intel Xeon Processor (Cascadelake)
Stepping:              6
CPU MHz:               2992.968
BogoMIPS:              5985.93
Hypervisor vendor:     KVM
Virtualization type:   full
L1d cache:             32K
L1i cache:             32K
L2 cache:              4096K
L3 cache:              16384K
NUMA node0 CPU(s):     0
Flags:                 fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 syscall nx
pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic m
ovbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch cpuid_fault invpcid_single ssbd ibrs i
bpb fsgsbase bmi1 hle avx2 smep bmi2 erms invpcid rtm avx512f avx512dq rdseed adx smap clflushopt clwb avx512cd avx512bw avx512vl xs
```

作业 3:

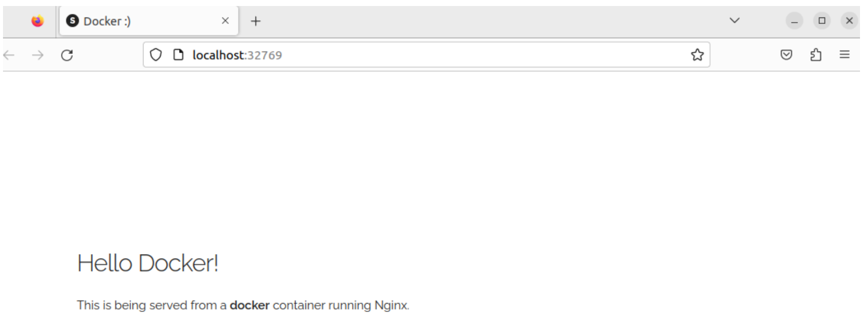
要先退出来刚刚作业 2 中的 container:

```
[root@77fc518c2de7 /]# exit
exit
[root@10-23-126-118 ~]#
```

然后在作业 3 之前有一些测试，我先跑了一遍:

```
[root@10-23-126-118 ~]# docker run --rm cloud_computing/static_site
Unable to find image 'cloud_computing/static_site:latest' locally
latest: Pulling from cloud_computing/static_site
5ab8e5104adc: Pull complete
4f4fb700ef54: Pull complete
12ba06104168: Pull complete
4fd24950bd05: Pull complete
0bc4824e250a: Pull complete
7b8d1d4a8910: Pull complete
d5341db39512: Pull complete
66c455cfe232: Pull complete
614b4c25101e: Pull complete
Digest: sha256:216f01f20cf3b436d9df1d850050cffa1151c0362e12d9292bf909c628ac66ab
Status: Downloaded newer image for cloud_computing/static_site:latest
Nginx is running...
^C[root@10-23-126-118 ~]# docker port static_site
Error response from daemon: No such container: static_site
[root@10-23-126-118 ~]# docker pull cloud_computing/static_site
Using default tag: latest
latest: Pulling from cloud_computing/static_site
Digest: sha256:216f01f20cf3b436d9df1d850050cffa1151c0362e12d9292bf909c628ac66ab
Status: Image is up to date for cloud_computing/static_site:latest
docker.io/cloud_computing/static_site:latest
```

```
[root@10-23-126-118 ~]# docker run -d -P --name static_site cloud_computing/static_site
dc1203119c12ca338d7cac0e7c85a6301d009da8049ba34b640e2645a4709eb4
[root@10-23-126-118 ~]# docker port static_site
80/tcp -> 0.0.0.0:32769
80/tcp -> [::]:32769
443/tcp -> 0.0.0.0:32768
443/tcp -> [::]:32768
```




```
[root@10-23-126-118 ~]# docker ps
CONTAINER ID   IMAGE                         COMMAND          CREATED
 NAMES
dc1203119c12   cloud_computing/static_site   "./wrapper.sh"   4 minutes a
, 0.0.0.0:32768->443/tcp, :::32768->443/tcp   static_site
```

```
[root@10-23-126-118 ~]# docker stop dc1203119c12
dc1203119c12
```

接下来就是这个题目的主要内容，我要使用-v、--rm 和-p 三个参数（将 container 的 80 端口映射到 host 的 8000 端口），创建 cloud_computing/static_site 容器。

```
[root@10-23-126-118 ~]# docker run --rm -d -p 8000:80 -v /var/html:/usr/share/nginx/html/mypage --name 10211900416 cloud_computing/s
tatic_site
cc8d682703d5bd39721da4e47c856b95612a8d582e10fad3abb787dbbf7b068b
```

localhost:8000/mypage/index.html

**Hello Docker!**

This is being served from a **docker** container running Nginx.

my id is 10211900416,my name is Tommy

作业 4：

```
[root@10-23-126-118 demo2]# docker build -t tommy/demo .
[+] Building 27.7s (9/9) FINISHED                                                                                        docker:default
 => [internal] load build definition from Dockerfile                                                                               0.0s
 => => transferring dockerfile: 529B                                                                                               0.0s
 => [internal] load .dockerignore                                                                                                  0.0s
 => => transferring context: 2B                                                                                                    0.0s
 => [internal] load metadata for docker.io/library/python:latest                                                                   0.4s
 => [1/4] FROM docker.io/library/python:latest@sha256:11454cd111137f54f2c91a624ab16b88d6dd31092a6f7fa267dea5e21b9f0a77            22.3s
 => => resolve docker.io/library/python:latest@sha256:11454cd111137f54f2c91a624ab16b88d6dd31092a6f7fa267dea5e21b9f0a77             0.0s
 => => sha256:42d620af35bebd96e135fdabc20b7d447d888901015956f8b231bba94d266442 8.05kB / 8.05kB                                     0.0s
 => => sha256:ed3d96a2798e8837be24597cabf44ce25585cb9db1d749299cb06d51349ea5c2 7.80MB / 7.80MB                                     0.2s
 => => sha256:11454cd111137f54f2c91a624ab16b88d6dd31092a6f7fa267dea5e21b9f0a77 2.22kB / 2.22kB                                     0.0s
 => => sha256:f1213685078145f6136360475dbaffd0f86dfe92133a7bc26d79602980b255dd 9.98MB / 9.98MB                                     0.3s
 => => sha256:5ae19949497e04289972756fe51cfac1a72b04fe2709e85a615945035c5a9a61 50.38MB / 50.38MB                                   0.8s
 => => sha256:6f18049a0455d5e717b580354b515d9ac661e1a28d6ca9d6f7bc85c0dd17a7cf 192.26MB / 192.26MB                                 4.8s
 => => sha256:1a9ad5d5550bdff7db4c3d035bf9550bcd1de06a7f178a26de1d082591a5b956 51.77MB / 51.77MB                                   4.8s
 => => extracting sha256:5ae19949497e04289972756fe51cfac1a72b04fe2709e85a615945035c5a9a61                                          5.5s
 => => sha256:ce39fa9d79d10239f5407e0bf43e2056f10cab5386d7319e220478fbed6a8323 5.79MB / 5.79MB                                     2.1s
 => => sha256:3a91ffcf88eadccd4b196138e3c783070af7e5239688128d2c7ed400289574e4 26.57MB / 26.57MB                                   5.8s
 => => sha256:ee82cc8e15068a2e57dec1805ebc11ae639bf10d967bf1b859794756916bd0de 235B / 235B                                         4.9s
 => => sha256:bf0dbf90a11587afe7b0b48c0f7bf182b2fb7823da8b298a45aa36977266369e 1.82MB / 1.82MB                                     5.0s
```

```
[root@10-23-126-118 demo2]# docker run -d -p 8080:5000 --name 10211900416 tommy/demo
06080553d6f94504b8e07ca653e44c338ea0688a0ae64813d264d60c962d415c
```

localhost:8080

hello world!